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Iron-Modified Biochar for Microplastic Removal from Agricultural Irrigation Water: A Critical Analysis of Published Data

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Background. Irrigation water is the most consistent route by which microplastics reach farmland soil and crops, and iron-modified biochar made from agricultural waste is widely proposed as a low-cost, magnetically recoverable adsorbent for intercepting them. The literature has grown tenfold in annual output since 2017 [9], which tends to consolidate assumptions faster than it tests them.

Methods. Rather than adding another material, this paper compiles quantitative values already reported — paired specific surface areas, points of zero charge, removal efficiencies and adsorption capacities, and irrigation water loadings — and records for each whether it was read from a retrieved full text or quoted at second hand. Three routine claims are tested against that compilation, and the circular-economy argument is tested by mass balance.

Findings. The isoelectric point of pH 6.59 commonly cited for polyethylene was measured in a trivalent aluminium coagulant and does not transfer to coagulant-free water, in which polyethylene and biochar alike carry net negative charge; electrostatic attraction therefore cannot account for capture under irrigation conditions, and physical entrapment is implicated instead. Reported removal efficiencies are not comparable between studies, clustering at 90–99 % for modified and unmodified materials alike while capacities for the same materials span a factor of 67, because removal percentage is governed largely by the dosage each investigator selected. A mass balance shows one hectare of paddy generates 2.4 to 60 times more husk-derived biochar than treating its own seasonal irrigation water would require — a real but bounded margin. Whether iron modification raises or lowers specific surface area remains open: it demonstrably does both, and the published record is too thinly sourced to locate the boundary.

Keywords rice-husk biochar · iron modification · microplastics · agricultural irrigation water · adsorption mechanism · evidence synthesis

1. Introduction

Plastic production continues to expand worldwide, and large quantities of debris enter natural environments every year. Mechanical abrasion, ultraviolet ageing and biological degradation break large items into fragments smaller than 5 mm, known as microplastics. Aquatic microplastic pollution has attracted attention for decades; terrestrial contamination, and in particular contamination of agricultural systems, remains comparatively underexamined.

Farmland receives microplastics through several pathways — plastic film residue, atmospheric deposition, sludge fertilisation and, most consistently, irrigation water. In many agricultural regions river water or reclaimed water is pumped directly onto fields. Surveys of irrigation water

and of the soils it reaches confirm both the presence and the accumulation of these particles [3]. Continuous irrigation introduces microplastics season after season, where they compact pore structure, reduce water retention, disturb microbial communities and impose oxidative stress on roots. Particles small enough to be taken up by roots translocate into stems and leaves, placing them on a path to the human diet [14]. Reported concentrations vary widely with catchment and method: surveys of irrigation systems give 3.1–64.3 particles L⁻¹ [3], while a river receiving urban discharge and used for vegetable irrigation reached 460–1953 particles L⁻¹ [15].

Practical treatment options for dispersed agricultural sites remain limited. Centralised advanced treatment requires capital and skilled operation that cannot cover numerous small plots, while simple sedimentation removes only the largest fragments. Biochar — a porous carbon

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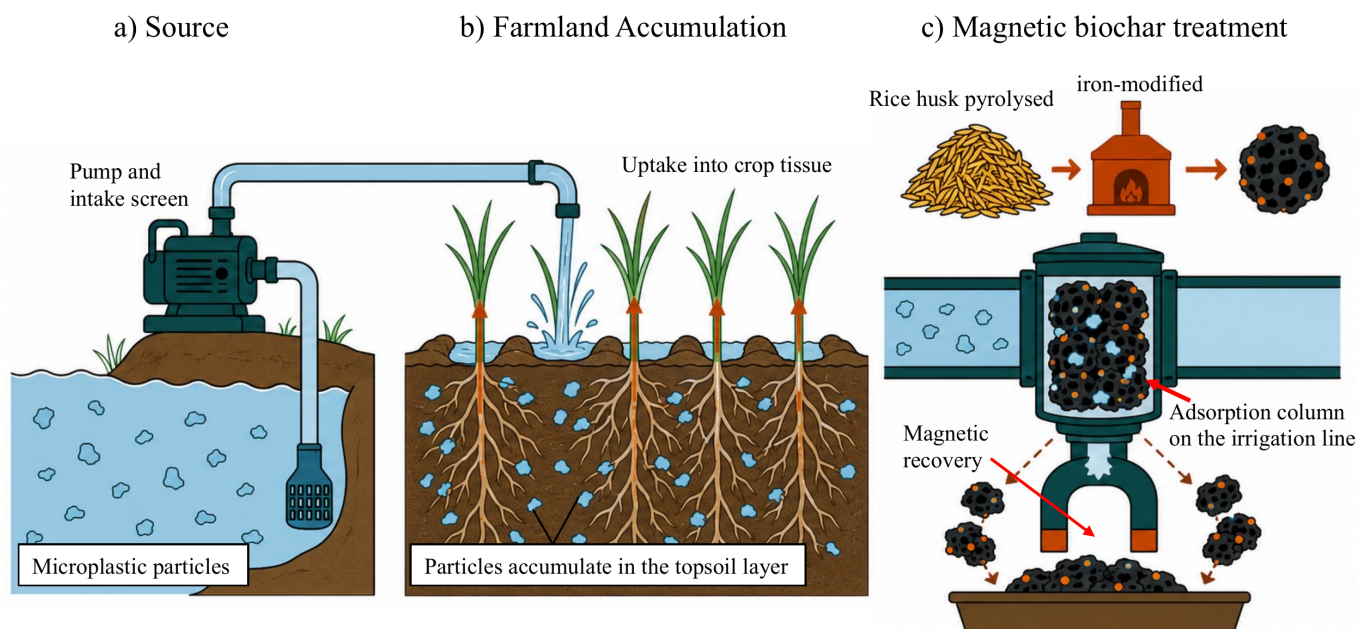


Figure 1. The system under review.

(a) Microplastics enter with pumped irrigation water; (b) particles accumulate in the topsoil and are taken up by crop tissue.

(c) Husk-derived iron-modified biochar is proposed to intercept them on the irrigation line and to be recovered magnetically.

produced by pyrolysing biomass under limited oxygen — has accordingly been examined as a low-cost adsorbent [1]. Rice husk is an attractive feedstock: it is generated in large quantities as a milling by-product and is otherwise burnt or dumped.

Pristine biochar has known drawbacks, among them modest surface area and the difficulty of recovering fine particles from treated water. Iron modification addresses both: iron oxide nanoparticles alter pore structure and surface chemistry, and render the composite magnetic, so spent adsorbent can be withdrawn with a magnet rather than filtered [2], and rice husk in particular has been used to make such materials [4]. Figure 1 shows the system these materials are intended to address. The preparation route typically followed is summarised in Figure 2.

The literature is now substantial: one bibliometric synthesis identifies 99 peer-reviewed articles published between 2017 and 2024, with a tenfold increase in annual output over that period [9]. Growth of that kind tends to consolidate assumptions faster than it tests them. This paper therefore does not add another material to the collection. It compiles quantitative values already reported, analyses them, and asks whether three claims made routinely in this literature survive that analysis: that iron modification increases specific surface area, that electrostatic attraction explains microplastic capture, and that reported removal efficiencies can be compared between studies. A fourth question, whether the circular-economy argument holds quantitatively, is addressed by mass balance.

2. Data compilation and analysis

2.1 Sources and evidence levels

Quantitative values were extracted from the primary literature and from one bibliometric synthesis of 99 articles [9].

Four categories were compiled: paired specific surface areas for biochars measured before and after iron modification; points of zero charge for rice-husk biochar and its iron-modified counterpart, together with reported isoelectric points for microplastic polymers; reported removal efficiencies and equilibrium adsorption capacities; and microplastic concentrations measured in agricultural irrigation water.

Each value carries an evidence level. Values read from a retrieved full text are distinguished throughout from values quoted at second hand in a review or synthesis, and the two are plotted with different symbols in Figure 3. Where a single research group reported the same material pair in more than one paper, the pair is counted once. The compiled datasets, the analysis script and a record of the evidence level attached to every value are available as described under Data availability.

2.2 Datasets

The paired surface-area dataset is given in Table 1. It spans parent biochars from 1.8 to 1330.7 m² g⁻¹. Four entries name a botanical feedstock — rice husk, peanut shell, corn straw and acacia wood — while three name a grade or a prior treatment rather than a feedstock, and iron is introduced by impregnation, co-precipitation or hydrothermal routes.

The performance dataset is given in Table 2. Removal efficiency and equilibrium adsorption capacity are reported separately, because as Section 3.3 shows they are not interchangeable. Capacity is defined as

$$q_e = (C_0 - C_e)V/m \quad (1)$$

and removal efficiency as

$$R(\%) = [(C_0 - C_e)/C_0] \times 100 \quad (2)$$

where C_0 and C_e (mg g^{-1} per litre) are the initial and equilibrium concentrations of the polymer in the liquid phase, V (L) is the suspension volume and m (g) is the mass of adsorbent.

2.3 Statistical treatment

Medians rather than means are reported for removal efficiency, because the distribution is bounded above at 100 % and strongly left-skewed. No imputation was performed; where a source did not report a quantity, that entry is absent rather than estimated.

No regression of the surface-area ratio on the parent surface area is reported. Such a fit is available and is included in the deposited script for transparency, but it is not evidence: the ratio $S_{\text{mod}}/S_{\text{raw}}$ carries S_{raw} in its denominator, so regressing it against S_{raw} returns a negative slope from uncorrelated noise alone. The apparent significance of such a fit is an artefact of its construction rather than a property of these materials, and reporting it would give the seven compiled pairs an authority the underlying data cannot support.

3. Results

3.1 The direction of the surface-area effect is not fixed

Whether iron modification raises or lowers specific surface area is not settled in this literature, and the compiled data of Table 1 shows why. Two of the seven pairs were read from a retrieved full text; the remaining five were quoted in review articles, and repeated searching failed to locate their primary sources. Values that cannot be traced cannot be checked by a reader, and are treated here as indicative only.

The two verified pairs point in opposite directions. Iron modification approximately doubled the surface area of a rice-husk char of low initial area, from 52.1 to $109 \text{ m}^2 \text{ g}^{-1}$ [5, 6], and reduced that of a commercial char of high initial

area by more than half, from 1330.7 to $525.11 \text{ m}^2 \text{ g}^{-1}$ [8]. Two measurements are enough to establish that the effect has no fixed direction. They are not enough to locate where it changes sign (Figure 3).

A mechanism consistent with both is that the iron oxide phase contributes surface roughness and secondary porosity when the parent char has little of either, and occupies existing pores when it has plenty. If that is right, a crossover exists somewhere between approximately 50 and $1300 \text{ m}^2 \text{ g}^{-1}$. The five secondary values are not inconsistent with it — every pair below $100 \text{ m}^2 \text{ g}^{-1}$ gained area, and the single pair that lost area began above $1300 \text{ m}^2 \text{ g}^{-1}$ — but agreement with values that cannot be verified is not evidence, and no threshold is proposed here.

Locating the crossover would require paired measurements on a series of parent chars spanning that range, prepared and measured under one protocol in one laboratory. No such series exists in the material compiled here, and assembling one from separate studies is precisely what the provenance problem above prevents. This is stated as a hypothesis available for testing, not as a result.

For rice husk the practical position is nonetheless clear. Rice husk pyrolysed at 500°C has been reported at $52.1 \text{ m}^2 \text{ g}^{-1}$ [5] and at $257.5 \text{ m}^2 \text{ g}^{-1}$ [12] — a fivefold spread at the same nominal temperature, which means nominal pyrolysis conditions do not determine where a char sits. Both values lie far below the commercial char that lost area on modification, so iron loading would be expected to increase surface area in either case. That expectation rests on one directly comparable measurement rather than on a fitted threshold, and should be treated accordingly.

3.2 The isoelectric point cited for polyethylene does not transfer to irrigation water

Electrostatic attraction between an adsorbent surface and a particle requires the two to carry opposite net charge,

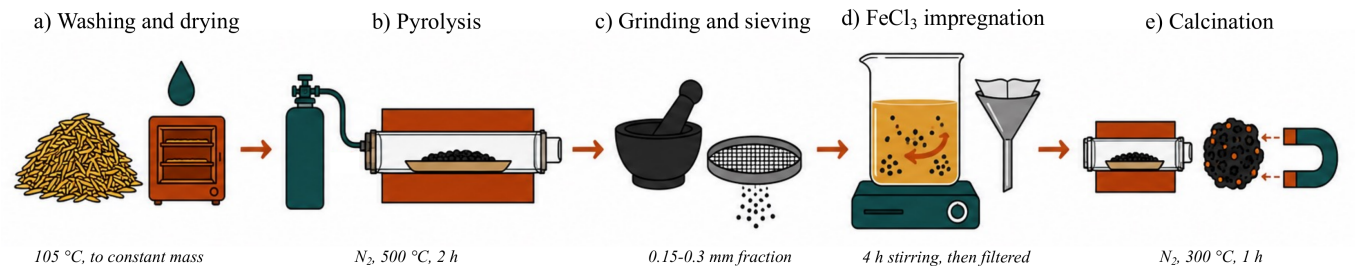


Figure 2. Preparation route typically used for iron-modified magnetic rice-husk biochar.

Husk is (a) washed and dried, (b) pyrolysed under nitrogen at around 500°C , and (c) ground and sieved.

It is then (d) impregnated with ferric chloride solution and filtered, and (e) calcined under nitrogen at a lower temperature to give the magnetic composite.

Table 1. Specific surface area of biochars reported before and after iron modification. Entries marked *review* were quoted in a synthesis rather than read from the original paper, and should be treated as provisional.

Feedstock	S_{raw} ($\text{m}^2 \text{ g}^{-1}$)	S_{mod} ($\text{m}^2 \text{ g}^{-1}$)	Ratio	Iron route	Evidence
Rice husk	52.1	109.0	2.09	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, hydrothermal	full text
Commercial biochar	1330.7	525.1	0.39	nano-Fe oxide impregnation	full text
Peanut shell	10.2	205.5	20.15	Fe loading, magnetic	review
Corn straw	93.7	313.9	3.35	Fe modification, magnetic	review
Acid-treated biochar	68.0	102.0	1.50	Fe impregnation	review
Acacia wood	322.0	394.0	1.22	Fe + Mg oxide impregnation	review
Generic biochar	1.8	19.8	11.00	Fe_3O_4 precipitation	review

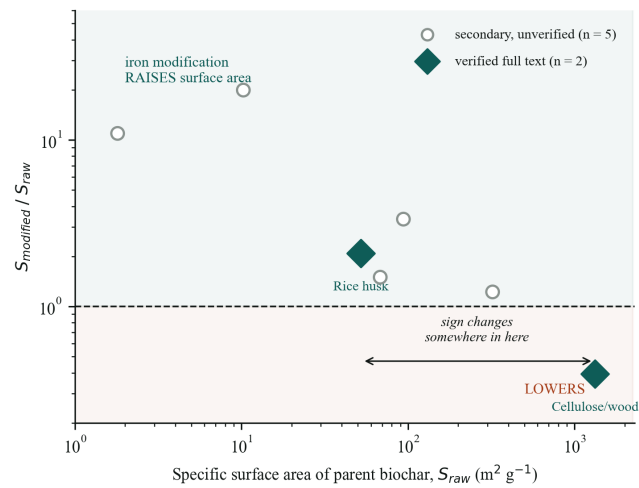


Figure 3. Ratio of modified to parent biochar specific surface area against the surface area of the parent biochar. The dashed line marks no change. Filled diamonds are the two pairs read from retrieved full texts, which lie on opposite sides of it; open circles are values quoted in review articles whose primary sources could not be located, shown for context only. No line is fitted: five of the seven values cannot be verified, and a regression of the ratio on the parent area would be negative by construction.

which occurs only between their respective points of zero charge. For rice-husk biochar the reported pH_{PZC} is 4.32 in 0.01 M NaCl, falling to 3.71 after iron modification [5]. The isoelectric point of polyethylene has been reported at pH 6.59 [10], which taken at face value would place an attraction window at pH 4.32–6.59 for raw biochar and pH 3.71–6.59 for the modified material — a widening of 26.9 % attributable to iron loading, and an apparently persuasive argument for modification.

That published isoelectric point, however, was measured in polyaluminium chloride solution, an electrolyte containing trivalent aluminium [10]. Trivalent cations adsorb strongly onto negatively charged surfaces and reverse their sign, producing an isoelectric point that does not exist in the absence of such an electrolyte. Polyethylene carries no ionisable surface groups; its charge in simple electrolytes arises from adsorbed hydroxide and chloride ions and becomes progressively more negative as pH rises, with no zero crossing between pH 3 and 10. Agricultural irrigation water is not dosed with coagulant, so the reported value cannot be carried across to it.

Both cases are shown in Figure 4. Where a trivalent coagulant is present, an acidic attraction window exists and iron modification widens it. In coagulant-free water, polyethylene and both biochars carry the same sign of charge from approximately pH 4 upward, and electrostatic forces oppose adsorption across the entire range relevant to irrigation rather than merely above pH 6.6.

Iron modification compounds this rather than relieving it. Lowering the point of zero charge from 4.32 to 3.71 makes the surface more negative at any circumneutral pH, and so marginally increases repulsion from a negatively charged particle. Whatever benefit iron modification confers must therefore arise from enlarged surface area and pore structure, and from magnetic recoverability, rather than from electrostatics — a conclusion opposite to the one

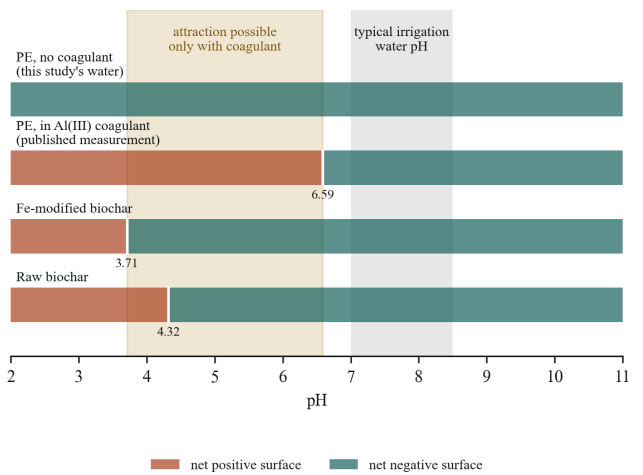


Figure 4. Net surface charge of raw rice-husk biochar, iron-modified rice-husk biochar and polyethylene as a function of pH, from published points of zero charge. Polyethylene is shown both as measured in a trivalent coagulant and as expected in coagulant-free water. The grey band marks the pH range of typical agricultural irrigation water.

Table 2. Reported microplastic removal efficiencies and adsorption capacities for biochar-based adsorbents. Dashes indicate the quantity was not reported in the source.

Polymer	Adsorbent	<i>R</i> (%)	<i>q</i> (mg g ⁻¹)
<i>Magnetic or metal-modified</i>			
PET microfibre	Nano-ferric loaded biochar	95.60	324.0
PHBV	MFe@BC iron oxide biochar	98.53	13.14
PS	Magnetic biochar	40.93	—
PS	MBC red mud/lignin	97.87	—
mixed	Magnetically modified	96.20	—
mixed	Mg/Zn magnetic biochar	97.14	—
PET microfibre	Nano-ferric loaded biochar	91.69	—
<i>Unmodified</i>			
PS	Banana peel biochar	92.16	—
PS	Dual-layer silica/corn straw	99.00	—
PE	Biochar	93.00	—
PE	Jujube waste biochar 700 rC	99.00	—
PE	Jujube waste biochar 300 rC	98.00	—
PS	London plane tree biochar 550 rC	—	60.05
PS	Amino-zeolite/H ₃ PO ₄	—	4.815

usually drawn from a lowered point of zero charge.

3.3 Reported removal efficiencies are not comparable between studies

The twelve published removal efficiencies compiled in Table 2 [9] span 40.9–99.0 %, with a median of 96.2 % for magnetic or metal-modified materials and 98.0 % for unmodified biochars. Taken at face value this ordering suggests that modification confers no benefit whatever. Adsorption capacities reported for the same class of materials, however, range from 4.82 to 324 mg g⁻¹ — a factor of 67 (Table 2, Figure 5).

The two metrics do not measure the same property. Removal efficiency is determined largely by the ratio of adsorbent dosage to initial concentration that each investigator selected: sufficient dosage drives any system above 90 %, and most published protocols use dosages generous enough to do so. The apparent parity between modified and unmodified materials is therefore an artefact of unmatched experimental conditions rather than a finding about the materials, and the near-universal 90–99 % band reported in this literature carries less information than its consistency suggests.

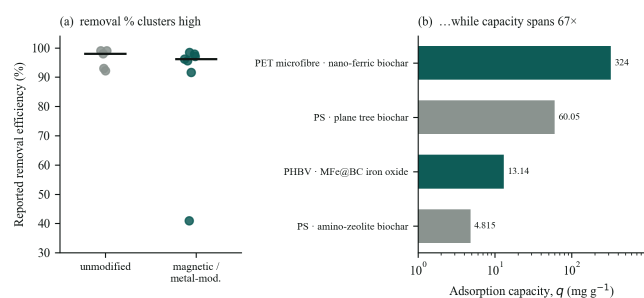


Figure 5. (a) Reported removal efficiencies for unmodified and magnetic or metal-modified biochars; horizontal bars are medians. (b) Reported adsorption capacities for the same class of materials, on a logarithmic axis. Efficiencies cluster within a narrow band while capacities span nearly two orders of magnitude.

Capacity is not a substitute unless it too is reported under matched conditions. Since q_e depends on C_0 through the adsorption isotherm, capacities measured at different initial concentrations are not directly comparable either. What a comparison requires is both materials measured in the same study, at the same dosage and the same C_0 , with q_e reported alongside the conditions used.

3.4 On-farm husk supply exceeds adsorbent demand, but not without limit

Taking a spherical polyethylene particle of 175 μm at a density of 0.94 g cm⁻³ gives a particle mass of 2.64 μg . At the survey mean of 64.3 particles L⁻¹ reported for irrigation water [3], and a seasonal paddy water input of 14 000 m³ ha⁻¹ [11], the microplastic load delivered to one hectare is 2.37 kg per season (Figure 6a). Because seasonal water input includes rainfall, this is deliberately an upper bound on the load arriving by irrigation.

The adsorbent this paper concerns is iron-modified, so demand is bounded by the iron-modified capacities of Table 2, which run from 13.14 to 324 mg g⁻¹. Meeting the load would require between 7.3 kg ha⁻¹ of adsorbent at the upper end of that range and 181 kg ha⁻¹ at the lower. The same hectare, at a paddy yield of 5.5 t ha⁻¹, produces roughly 1.1 t of husk and hence about 440 kg of biochar at a 40% pyrolysis yield, which lies within the 33–42% range measured for rice husk between 300 and 700 °C [12]. Supply exceeds demand by a factor of 2.4 to 60 (Figure 6b). Under the least favourable combination within that range — the highest measured water concentration against the lowest iron-modified capacity — demand rises to 256 kg ha⁻¹ against 440 kg ha⁻¹ of supply, a ratio of 1.7.

That bound depends on the restriction to iron-modified materials, and the restriction should be stated rather than assumed. The lowest capacity anywhere in Table 2 is 4.815 mg g⁻¹, reported for an unmodified amino-zeolite biochar. An adsorbent performing at that level would require 493 kg ha⁻¹ against 440 kg ha⁻¹ of supply, and the margin would close entirely. The circular-economy case made here is therefore a case for the modified material specifically, not for husk-derived biochar in general.

The margin is real but bounded, and the bound is the useful result. A reserve of between two and sixty fold means that factors eroding working capacity are not peripheral to the concept but central to it: competition from dissolved or-

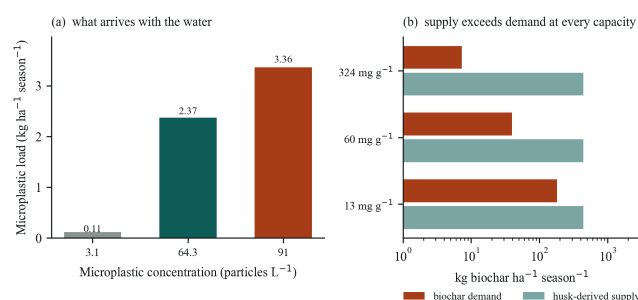


Figure 6. (a) Microplastic mass delivered to one hectare per season at three reported irrigation water concentrations, assuming 14 000 m³ ha⁻¹ and 175 μm polyethylene spheres. (b) Adsorbent demand at three published iron-modified capacities against the biochar obtainable from the same hectare's rice husk, on a logarithmic axis.

ganic matter, failure to reach equilibrium in a flowing contactor, and capacity loss across reuse cycles all consume it directly. This calculation assumes equilibrium capture of every particle, which no real contactor achieves, so the true margin is narrower than the figures above.

4. Discussion

4.1 Implications for the capture mechanism

The surface-charge result of Section 3.2 constrains which mechanisms can operate under irrigation conditions. Physical entrapment within the pore network does not depend on the sign of the surface charge and remains available at any pH; so do surface complexation at iron oxide sites and van der Waals attraction. Electrostatic attraction, by contrast, is available only below the point of zero charge of the adsorbent, at roughly pH 3.7 for the modified material, and is repulsive above it. These pathways, and the way dissolved humic acid interferes with them, are summarised in Figure 7.

Two published studies of iron-modified biochars report removal held effectively flat across pH 4–9, above 91.7% and above 95% respectively [7, 8]. Sustained capture across a range in which electrostatics are unfavourable throughout is difficult to attribute to charge attraction, and is what pore-filling and dispersive interactions would predict. Independent column work points the same way: peanut-shell biochar and its MgO-modified form raised microplastic retention in saturated porous media from 34.2% to 59.1% and 75.5% respectively in the absence of humic acid, with retention falling once humic acid was introduced [13] — a pattern consistent with physical capture that dissolved organic matter obstructs, rather than with charge attraction. Where electrostatic attraction is invoked in this literature it is generally inferred from a lowered point of zero charge rather than demonstrated, and Section 3.2 shows that inference runs the wrong way.

4.2 Implications for how these materials are made

If the benefit of iron modification lies in pore structure rather than surface charge, then preparation should be optimised for the former. Section 3.1 shows this is not simply a matter of maximising surface area at every stage: iron loading can reduce the surface area of a char that already has a great deal, so a parent char produced at high pyrolysis

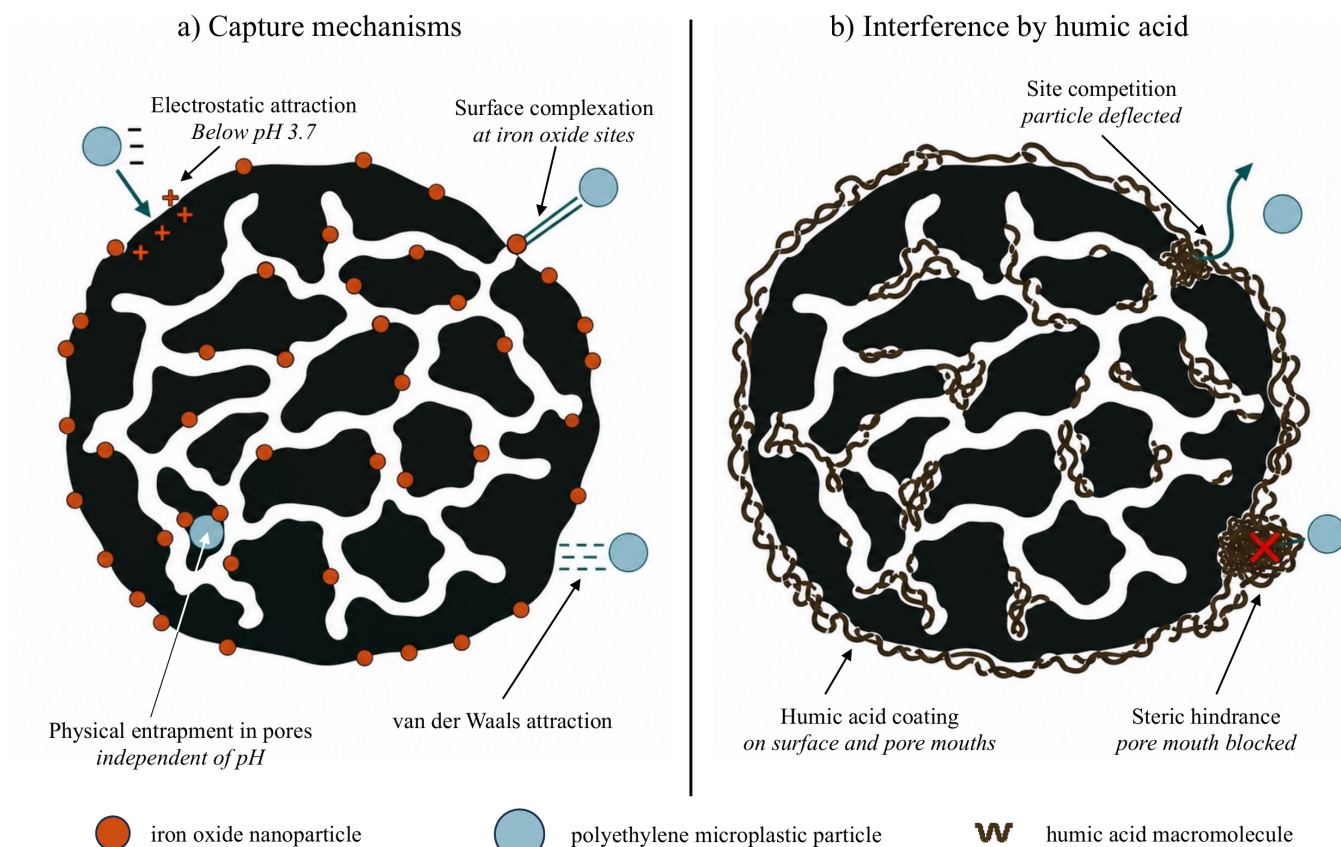


Figure 7. Mechanisms proposed for microplastic capture by iron-modified biochar.

(a) Physical entrapment within the pore network, independent of pH; electrostatic attraction, attractive only below the adsorbent's point of zero charge and repulsive above it; surface complexation at iron oxide sites; and van der Waals attraction.

(b) Dissolved humic acid coats the external surface and pore mouths, competing for sites and imposing steric hindrance on approaching particles.

temperature may leave modification less room to help, and possibly none. For feedstocks and temperatures yielding parent chars in the tens of $\text{m}^2 \text{g}^{-1}$, of which rice husk pyrolysed at 500 °C is one, the single directly comparable measurement available suggests modification roughly doubles the available surface. The combination of feedstock and pyrolysis temperature should therefore be chosen with the subsequent modification step in view rather than independently of it — though quantifying that trade-off requires the systematic series identified in Section 3.1, which does not yet exist.

4.3 Implications for how studies should report their results

Section 3.3 identifies a reporting problem rather than a scientific one, but it limits what the accumulated literature can support. Three practices would make future results comparable. First, equilibrium adsorption capacity should be reported as the primary metric, with removal efficiency given alongside the dosage and initial concentration that produced it. Second, modified and unmodified materials should be compared within a single study at matched dosage and matched initial concentration; comparisons assembled across studies cannot be interpreted. Third, the electrolyte composition used for surface-charge measurements should be stated explicitly, since Section 3.2 shows that a value measured in a coagulant does not describe the same material in natural water.

4.4 Limitations of this analysis

The surface-area comparison is the weakest part of this analysis, and is presented as an open question rather than a finding for that reason. Only two of the seven compiled pairs could be traced to a retrieved full text; the other five were quoted in review articles whose primary sources repeated searching failed to locate. Recovering those five, or better, generating a purpose-built series under one protocol, would be the single most valuable extension of this work. The expectation that polyethylene lacks an isoelectric point in coagulant-free water follows from its surface chemistry and from reported zeta-potential trends, but was not measured here, and a direct measurement in a simple electrolyte would settle Section 3.2 conclusively. Published capacities are dominated by PET, PHBV and polystyrene rather than polyethylene, so the range quoted in Section 3.4 is transferred across polymers. The mass balance assumes equilibrium capture and takes seasonal water input as an upper bound on irrigation volume.

This analysis is also confined to what has been published. Negative results — modifications that reduced performance, materials that failed to magnetise sufficiently for recovery — are systematically less likely to appear in the literature, so the compiled removal efficiencies should be read as an upper envelope of what these materials achieve rather than as a representative sample. Nothing

here demonstrates that iron-modified rice-husk biochar removes microplastics from real irrigation water; that requires experiments this analysis was written to inform.

5. Conclusion

Reanalysis of published data on iron-modified biochar for microplastic removal yields three results and one open question. The isoelectric point commonly cited for polyethylene was measured in a trivalent coagulant and does not describe the polymer in irrigation water, where adsorbent and particle alike carry negative charge and electrostatic attraction is unavailable at any relevant pH. Reported removal efficiencies are not comparable between studies, because they are governed largely by the dosage each investigator selected. And a hectare of paddy generates between 2.4 and 60 times more husk-derived biochar than would be needed to treat its own irrigation water using an iron-modified adsorbent, a margin that is real but narrow enough that losses to organic matter competition and reuse matter. The open question is whether iron modification helps or harms specific surface area: it demonstrably does both, depending on the parent char, but the published record is too thinly sourced to say where the boundary lies.

Taken together these results do not argue against husk-derived magnetic biochar as a candidate adsorbent. They argue that the evidence currently offered in its support is weaker than its volume suggests, and they identify what would strengthen it: capacities measured at matched dosage, surface-charge values reported with their electrolyte, and parent-char surface areas chosen with the modification step in view. The circular-economy case, which has usually been made in principle, can also be made in numbers — and is better for being bounded.

Declarations

Funding. This research received no funding.

Conflict of interest. The author declares no competing financial or non-financial interests.

Ethics approval. Not applicable. This study analyses previously published data and involved no human participants, human tissue or animals.

Author contributions. J.K. is the sole author and carried out the work unaided. J.K. conceived the study, defined its scope and research questions, compiled and curated the dataset, directed and verified the analysis, produced the figures, wrote and revised the manuscript, and approved the final version.

Data availability. All quantitative values analysed in this study were extracted from previously published sources, each cited in the reference list. The compiled datasets — paired specific surface areas before and after iron modification, surface charge values, reported removal efficiencies and adsorption capacities, and irrigation water loading parameters — together with a source record documenting the origin and evidence level of every value, and the analysis script reproducing the mass balance of Section 3.4 and the quantities plotted in Figures 3–6, are openly available from Zenodo at <https://doi.org/10.5281/zenodo.22130291> under a CC BY 4.0 licence. No new experimental data were generated.

Use of generative artificial intelligence. Claude (Anthropic) was used to compile published values, run the statistical analysis reported in Section 3, generate Figures 3–6, and draft manuscript text. The author designed the study, verified all analysis and sources, and takes full responsibility for the work. Claude is not an author.

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